Errata and Counterexamples for *Approximate Majority Analyses using Tri-molecular Chemical Reaction Networks*

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found while formalizing the paper in Lean 4 / Mathlib

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Abstract

These errata were found while producing a machine-checked formalization of the paper in Lean 4 with Mathlib. Each entry identifies the exact printed passage, gives either a short quotation or an exact mathematical transcription, and states the correction; where a corrected theorem has been machine-checked we name the corresponding Lean declaration. Every location below was checked against the published Springer PDF of *Natural Computing* 19 (2020), 249–270, DOI `10.1007/s11047-019-09756-4`. We give both the printed publication page and the one-based PDF page, counted from the article's title page.

The paper's entry and consensus conclusions are not refuted by this audit. One additional persistence clause — Theorem 4's preservation of relaxed consensus for the following n^γ interactions — is false as printed. The remaining items are proof gaps, omitted hypotheses or qualifiers, and typographical errors. We do *not* claim that every downstream use survives: the false preservation clause has no corrected downstream proof at its printed strength. The full-range tail of Lemma 12 / Theorem 5 is true, but the printed proof does not establish it; a replacement state-dependent proof has been machine-checked.

Classification scheme. (E) a printed statement is false; (P) a genuine proof erratum with the statement still true; (U) an assumption, scope condition, probability qualifier, or stronger conclusion is under-stated; (T) a typo or typesetting omission with unambiguous intended meaning; and (O) *open after audit* — the printed proof does not establish the stated parameter range, and no counterexample or replacement proof is currently known.

The (O) label exists because E/P/U/T do not faithfully cover a claim whose proof fails while its statement is neither proved nor refuted: labelling such a claim (P) would, by the definition above, assert that the statement is known true. Exactly one defect is (E) — Theorem 4's preservation clause, recorded first below and refuted by an explicit fixed-adversary family. No defect remains in class (O): the former open case, the full-range Lemma 12 / Theorem 5 tail, is now classified (P). The replacement proof retains each competitor's actual initial gap before summing the failure probabilities and yields the canonical Lean declarations `Tri.Multi.lemma12` and `Tri.Multi.theorem5`.

Notation

Throughout, n is the population size, x and y are the counts of the two species with $x + y = n$, $\gamma \geq 1$ is the paper's confidence parameter, and lg is the base-2 logarithm. A reaction event is *productive* if it changes the counts.

Erratum 1 (Section 4.4 — `creation-noise tolerance (P)`). Location. *Natural Computing* 19 (2020), p. 259 (Springer PDF p. 11), Sect. 4.4, first paragraph: the choice of the tolerance used to control the imbalance between the two half-rate creations.

As printed. The argument fixes the tolerance

$$d \;=\; \frac{\gamma n \lg n}{8\Delta}$$

and asserts creation-noise control at $\exp(-\Theta(\gamma \lg n))$ uniformly on the stated range $\sqrt{\gamma n \lg n}/2 < \Delta < n$.

Issue. Put $g = \gamma \lg n$. For the displayed tolerance and a creation budget $H = 15n$, the Chernoff exponent is

$$\frac{d^2}{2H} = \frac{g^2 n}{1920\Delta^2}.$$

This has order g when Δ has order $\sqrt{gn}$, but it is not $\Omega(g)$ uniformly on the printed range. A small fixed coefficient of g is consistent with the claimed asymptotic order; the defect is the loss of that order as Δ grows relative to $\sqrt{gn}$.

Why. At the lower boundary $\Delta = \sqrt{gn}/2$, direct substitution gives

$$\frac{d^2}{2H} = \frac{g}{480} = \Theta(g).$$

The boundary itself is excluded by the printed strict inequality, but for every fixed $\kappa > 1/2$, taking $\Delta = \kappa\sqrt{gn}$ gives the same order, namely $g/(1920\kappa^2)$. In contrast, for $\Delta = cn$ with fixed $0 < c < 1$ and $g/n \to 0$,

$$\frac{d^2}{2H} = \frac{g^2}{1920c^2 n} = o(g).$$

The condition $g/n \to 0$ holds, in particular, for fixed γ as $n \to \infty$; these choices of Δ eventually lie in the printed range. Hence this tolerance and budget do not supply a uniform exponent of order g over that range.

At $n = 2^{20}$ and $\gamma = 1$ (so $g = 20$), the corresponding numerical values are:

Δ	d	$d^2/(2 \cdot 15n)$
$\sqrt{\gamma n \lg n}/2$	1144.9	4.17×10^{-2}
$n/16$	40.0	5.09×10^{-5}
$n/4$	10.0	3.18×10^{-6}
$n/2$	5.0	7.95×10^{-7}

The first row is the valid fixed-coefficient exponent $g/480$; the large-gap family above establishes the failure of uniformity.

Fix. Use stage-specific creation budgets and deviations rather than a single global tolerance tied to Δ. The formalization does exactly that: `Tri.theorem2_singleB` carries a per-stage (D, H) pair, and the underlying concentration lemma is in fact *stronger* in temporal scope than the printed fixed-count step — `Tri/SingleBCreationTailY.lean` proves the maximal stopped bound

$$\Pr\big[\exists t:\; C_t \le H \text{ and } CY_t - CX_t \ge D\big] \le \exp\!\left(-\frac{D^2}{2H}\right)$$

with an explicit constant, where the paper states only an asymptotic $\exp(-\Theta(\gamma \lg n))$. The theorem-level conclusion of the Single-B clause is unaffected.

Erratum 2 (Theorem 4 — `preservation clause (E)`). Location. *Natural Computing* 19 (2020), pp. 261–262 (Springer PDF pp. 13–14), Theorem 4's second conclusion and its preservation argument for the subsequent n^γ interaction events.

As printed. Under $\Delta_0 := 2x - n \geq \sqrt{\gamma n \lg n}$ and $z < \Delta_0/16$, with probability $1 - \exp(-\Omega(\gamma \lg n))$ a relaxed X-consensus is reached within $O(\gamma n \lg n)$ interactions *and is preserved for the subsequent n^γ interaction events.*

Why it is false. The dynamics do not depend on the confidence parameter γ, whereas the window length n^γ does. With the Byzantine count z a fixed constant, escaping the buffer costs only $n^{-O(z)}$ per attempt, and a window of unbounded polynomial degree overwhelms it. Concretely, take

$$z = 2, \qquad \gamma = 17, \qquad (x, y, z) = (n - 16,\, 14,\, 2),$$

which lies exactly on the printed relaxed-consensus boundary $x = n - 8z$ and satisfies both hypotheses for all large n (here $\Delta_0 = n - 32$, and $(n-32)^2 \geq 17\, n \lg n$ and $2 < (n-32)/16$).

Use the paper's own declared worst response, in which Byzantine molecules always favour Y; this is a *fixed, non-adaptive* adversary, so allowing adaptive adversaries cannot rescue the claim. Partition the raw timeline into blocks of $29n^2$ interactions. From any state with $0 \leq y \leq 14$, one block drives the process to $y = 15$ — i.e. out of the relaxed guard — with probability at least $c\, n^{-14}$, where $c = 1/(2 \cdot 4^{14} \cdot 6^{14})$: force y down to 0; nucleate $y = 1$ from a triple $\{X, B, B\}$ whose two Byzantine molecules both present Y, so that the reaction reads $X + Y + Y \to 3Y$ and a genuine X becomes a genuine Y; then take fourteen prescribed up-steps, each costing at most a factor $1/(6n)$. There are $\Theta(n^{15})$ such blocks in n^{17} interactions, so

$$\Pr[\text{no exit by } n^{17}] \;\leq\; \left(1 - c\, n^{-14}\right)^{\Theta(n^{15})} \;\leq\; \exp(-\Theta(n)) \;\longrightarrow\; 0,$$

whereas the theorem claims an exit probability of $\exp(-\Omega(\gamma \lg n)) \to 0$.

Scale, stated faithfully. The constant is small: the crossover, where $cn/29$ first exceeds 1, is near $n \approx 1.2 \times 10^{21}$. Since the theorem is asymptotic this is a legitimate refutation, and the crossover lies far below the n_0 at which this development's other headline theorems are even asserted.

Consequence. Theorem 4's *entry* conclusion is unaffected. The preservation conclusion requires either a window tied to the Byzantine count (rather than an unbounded polynomial degree), or a buffered guard with slack growing in γ. The Lean development records the printed clause in `Tri/Theorem4Statement.lean` as known-false and does not attempt it; the corrected reached-by-deadline entry clause is machine-checked as `Tri.Byzantine.theorem4_entry : Theorem4_entry_statement`, with failure at most $n^{-\gamma/136}$.

Related printed slips in the same passage. The definition of relaxed consensus by the *equality* $x = n - 8z$ cannot be preserved under any dynamics and must be read as $x \geq n - 8z$; the proof's "$x \geq n - 2z$ is equivalent to $y < z$" should read $y \leq z$; and "two doublings" from $M = 2z$ reaches the boundary $M = 8z$, so strict exit needs one further upward step.

Erratum 3 (Lemma 4 — `proof`). Location. *Natural Computing* 19 (2020), p. 256 (Springer PDF p. 8), Lemma 4 proof, first displayed lower bound on the productive-reaction success probability.

As printed. After setting $d = \gamma \lg n$ and restricting to the process before the first time $y > n/k + d$, the paper prints

$$\frac{x}{n} \;\geq\; 1 - \frac{n - kd}{kn} \;>\; 1 - \frac{1}{k},$$

and concludes that each productive reaction succeeds with probability $p > 1 - 1/k$.

Issue. Both the displayed inequality and its conclusion fail. From $y \leq n/k + d$ one gets $x = n - y \geq n - n/k - d$, hence

$$\frac{x}{n} \;\geq\; 1 - \frac{1}{k} - \frac{d}{n} \;=\; 1 - \frac{n + kd}{kn},$$

so the numerator is $n + kd$, not $n - kd$: the printed line overstates x/n by $2d/n$. With the correct sign the bound reads $1 - 1/k - d/n < 1 - 1/k$, so the conclusion $p > 1 - 1/k$ does not follow, and it is genuinely false at states inside the band the proof claims to cover. The exact per-productive-event success probability

$$p_y = \frac{\binom{x}{2} y}{\binom{x}{2} y + x \binom{y}{2}} = \frac{x - 1}{n - 2}$$

does not repair it either. Numerical witnesses at the band's upper edge $y = n/k + d$:

k	n	d	p_y	$1 - 1/k$
4	10^3	50	0.7004	0.7500
8	10^3	50	0.8257	0.8750
6	4096	60	0.8190	0.8333
4	10^6	20	0.74998	0.75000

In each case $p_y < 1 - 1/k$, so the claimed uniform floor does not hold on the band.

Consequence. Lemma 4's conclusion is nevertheless true. The formalization (`Tri/PaperLemma4.lean`) proves it by a corrected stopped argument: with $P = n/k$ the initial minority it uses the smaller safety buffer $B = \lfloor P/8 \rfloor$, on whose band the exact productive kernel does have a uniform adaptive success floor; a Feller bound controls escape from the band and an adapted Chernoff bound controls failure to reach consensus by the paper's productive deadline. So this is a defect of the printed proof, not of the statement.

Erratum 4 (Corollary 2 — `statement`). Location. *Natural Computing* 19 (2020), p. 256 (Springer PDF p. 8), displayed statement of Corollary 2.

As printed. "Since stage s of phase 2 starts with $y \leq n/2^s \leq n/6$ and ends with $y = \min\{n/2^{s+1},\ \gamma \lg n\}$, it completes properly, within at most $3n/2^s$ productive reaction events, with probability at least $1 - \exp(-\Theta(\gamma \lg n))$."

Issue. The min should be a max, interpreted as a stopping threshold. The issue concerns the first stopped endpoint, not whether a smaller value can occur later on an unrestricted trajectory.

Why. The paper's own definition of a phase-2 stage (L825) is unambiguous:

> "A typical stage s in phase 2 starts with $y \leq n/2^s$ and continues while $n/2^{s-1} \geq y > n/2^{s+1}$. It completes properly if $y \leq n/2^{s+1}$."

The stage's own target is $y \le n/2^{s+1}$, while phase 2 as a whole terminates at its floor $y \le \gamma \lg n$ (L811). Writing $A = n/2^{s+1}$ and $B = \gamma \lg n$, the union of these two lower stopping sets is exactly

$$\{y \le A\} \cup \{y \le B\} = \{y \le \max\{A, B\}\}.$$

For every trajectory, the first entrance into either lower stopping set is therefore the first entrance into the set on the right. This argument does not require pathwise monotonicity of y: a productive reaction $X + Y + Y \to 3Y$ increases y by one.

Fix. Replace min by max and read the combined lower stopping condition as

$$y \le \max\{n/2^{s+1}, \gamma \lg n\}.$$

Since y is integer-valued, exact equality with the displayed threshold need not be possible. During the live part of the combined stage, both $y > A$ and $y > B$ hold, equivalently $y > \max\{A, B\}$.

The Lean development sidesteps the issue entirely by formalizing the *stage definition* (L825), which involves no min or max: writing $P = n/2^s$ and $B = n/2^{s+1}$, the stage runs from $y \le P$, escapes if $y > 2P$, and completes at $y \le B$, within $3P$ productive events.

Erratum 5 (Lemma 6 — `threshold`). Location. *Natural Computing* 19 (2020), p. 260 (Springer PDF p. 12), displayed statement of Lemma 6.

As printed. Under the unequal-rate hypothesis, the gap "increases to $\min\{2\Delta, n\}$" within $5n$ productive reaction events.

Issue. Classification: (T). Since $x + y = n$, the signed gap satisfies

$$G = x - y = 2x - n \equiv n \pmod 2.$$

Productive reactions change G by $+2$ or -2, so every reachable gap has the same parity as n. Thus the displayed target cannot be read as exact equality in every population size. A concrete witness is $n = 9$, initial split $(x, y) = (5, 4)$, and $\Delta = 1$: all reachable gaps are odd, so the literal target 2 is unreachable. Downward moves are allowed; for example, one reaction $X + Y + Y \to 3Y$ gives $(x, y) = (4, 5)$ and gap -1.

Fix. Read the conclusion as reaching the first reachable gap at least the threshold, i.e. $G_t \ge \min\{2\Delta, n\}$. This is the intended threshold statement and is what the proof supports.

Erratum 6 (Lemma 6 — `ratio step (P)`). Location. *Natural Computing* 19 (2020), p. 260 (Springer PDF p. 12), Lemma 6 proof, the comparison of the safety ratio $2n/(2n + \Delta)$.

As printed.

$$\frac{2n}{2n + \Delta} < 1 - \frac{\Delta}{2n}.$$

Issue. Classification: (P). The inequality has the wrong direction. For $n > 0$ and $\Delta > 0$,

$$\frac{2n}{2n + \Delta} - \left(1 - \frac{\Delta}{2n}\right) = \frac{\Delta^2}{2n(2n + \Delta)} > 0.$$

Fix. Use the valid exponential estimate

$$q := \frac{2n}{2n + \Delta} = 1 - \frac{\Delta}{2n + \Delta} \le \exp\left(-\frac{\Delta}{2n + \Delta}\right).$$

Rounding the quarter-gap barrier upward gives

$$q^{\lceil \Delta/4 \rceil} \le \exp\left(-\frac{\lceil \Delta/4 \rceil \Delta}{2n+\Delta}\right) \le \exp\left(-\frac{\Delta^2}{8n+4\Delta}\right).$$

For $0 < \Delta < n$, the exponent is at least $\Delta^2/(12n)$, so this retains the required order of the safety estimate.

The last bound is machine-checked as `Tri.lemma6Safety_power_exp` in `Tri/PaperLemma6.lean`. It is the safety term in `Tri.lemma6Trace_failure_explicit`, whose full bound for the stopped trace after $5n$ slots is

$$\exp\left(-\frac{\Delta^2}{8n+4\Delta}\right) + \exp\left(-\frac{\Delta^2}{80n+16\Delta}\right).$$

These probability results use the fixed-rate model of `Tri.lemma6_paper`; extending that formal statement to arbitrary history-dependent rates is a separate formalization obligation.

Erratum 7 (Lemma 12 — `gap (P)`). Location. *Natural Computing* 19 (2020), p. 263 (Springer PDF p. 15), Lemma 12 proof, the type-(a) estimate immediately after the four stopping alternatives.

As printed (statement). "Consider a point in the computation where $x - z_{\max} = \Delta \ge \sqrt{\gamma n \lg n}$. Let $m \le n/(\gamma \lg n)$. Then $x - z_{\max} \ge \min\{2\Delta,\ n\}$ within $\Theta(n)$ productive reaction events, with probability $1 - \exp(-\Omega(\gamma \lg n))$."

As printed (the step at issue, in the proof). "By Lemma 11, completion of type (a) occurs with probability $m \exp(-\Omega(\gamma \lg n)) = \exp(-\Omega(\gamma \lg n))$ since $m < n$."

Issue. Classification: (P). The displayed absorption of the factor m into the $\Omega(\cdot)$ is not justified by $m < n$ alone. It additionally requires the constant hidden in the exponent to exceed $\ln 2$, which the paper never establishes and which the natural constants coming out of the argument do not satisfy.

Why. Write the exponent explicitly: Lemma 11 supplies a failure mass of the form $m \exp(-c\,\gamma \lg n)$ for some constant $c > 0$ determined by the argument. Using only $m \le n = 2^{\lg n}$,

$$m \exp(-c\,\gamma \lg n) \ \le\ \exp\big((\ln 2)\lg n - c\,\gamma \lg n\big) \ =\ \exp\big(-(c\gamma - \ln 2)\lg n\big).$$

For this to be $\exp(-\Omega(\gamma \lg n))$ one needs $c\gamma - \ln 2 \ge c'\gamma$ for some $c' > 0$, i.e.

$$\gamma\,(c - c') \ \ge\ \ln 2.$$

Since the lemma is asserted "for any constant $\gamma \ge 1$", this forces $c > \ln 2 \approx 0.693$. The constant that the machine-checked version of this argument actually produces is $c = 1/82944$, so for that constant the absorption fails outright for every γ in a very large range. The paper's own hypothesis $m \le n/(\gamma \lg n)$ helps only by a log factor and does not change the conclusion.

Fix. The earlier formal wrapper exposed the exact additional condition required by the printed coarse union-bound route:

$$288m + 144 \ \le\ \exp\left(\frac{\gamma \lg n}{165888}\right),$$

but the replacement proof does not use that route. For a competitor with initial gap g_Y, it retains both the Gaussian decay at the minimum gap Δ and the additional decay proportional to $g_Y - \Delta$. Competitors with population comparable to x_0 are then counted by total population rather than by m, while smaller competitors are absorbed by the additional decay. The resulting 144-substage error is at most

$$432 \exp\left(-\frac{\gamma \lg n}{663552}\right)$$

throughout the printed range $m\gamma \lg n \le n$ and $\gamma n \lg n \le \Delta^2$. This is `Tri.Multi.lemma12`.

Remark. The printed statement is true. The error is in discarding the competitor-dependent decay before summing and then trying to absorb the resulting artificial factor m.

Erratum 8 (Lemma 12 — `proof`). Location. *Natural Computing* 19 (2020), p. 264 (Springer PDF p. 16), Lemma 12 proof, the paragraph beginning "Next, consider completions of type (c)," where progress against a fixed species is followed by an invocation of Lemma 11.

As printed. The proof first obtains, for a fixed species Y, a new large X–Y gap, and then feeds that conclusion into Lemma 11.

Issue. Classification: (P). Lemma 11 assumes a *global* plurality gap $x - z_{\max} \ge \Delta'$. A large gap against one fixed Y at that species' individual hitting time does not by itself establish the global premise against every competitor. This is logically separate from the species-factor absorption error above.

Fix. The replacement proof uses one synchronized global stopping process. Its safety and progress potentials are summed over all competitors while retaining each actual initial pair gap. Consequently every terminal failure is charged once, and no fixed-Y conclusion is passed to the global Lemma 11. This proof is implemented by `productiveProperStage_progress_state`, `productiveProperStageStateError_le_exp`, and the public `lemma12`.

Erratum 9 (Lemma 12 — `terminal line`). Location. *Natural Computing* 19 (2020), p. 264 (Springer PDF p. 16), final sentence of the Lemma 12 proof.

As printed. The proof ends with the uncapped conclusion $x - z_{\max} \ge 2\Delta$.

Issue. Classification: (T). The uncapped inequality is impossible when $\Delta > n/2$. This is a terminal typesetting omission, since the lemma's own statement already uses the capped target.

Fix. The final line should read $x - z_{\max} \ge \min\{2\Delta, n\}$, with the terminal branch interpreted as consensus.

Erratum 10 (Lemma 13 — `exponent`). Location. *Natural Computing* 19 (2020), p. 264 (Springer PDF p. 16), final sentence of the Lemma 13 proof.

As printed. The proof says that λmn raw interactions, with a productive-reaction floor of order $\Theta(1/m)$, give failure $\exp(-\Theta(n/m))$.

Issue. Classification: (U). The Chernoff mean for the productive count is actually $\Theta(n)$, so the constant-fraction lower-tail calculation gives $\exp(-\Theta(n))$. The printed exponent understates the calculation.

Fix. State the stronger $\exp(-\Theta(n))$ clock tail. The weaker reading $\exp(-\Omega(n/m))$ is harmless for the downstream use.

Erratum 11 (Theorem 3 — `parameter algebra (P)`). Location. *Natural Computing* 19 (2020), p. 261 (Springer PDF p. 13), proof of Theorem 3, parameter-algebra paragraph immediately following the theorem statement.

As printed. With

$$c = \frac{(1+\xi)(1-\alpha)}{1+\alpha}, \qquad 0 < \xi < \alpha < 1,$$

the proof asserts the intermediate inequality $c < \alpha$.

Issue. Classification: (P). This inequality is false. Taking $\alpha = 1/10$ and $\xi = 1/20$ satisfies all the displayed assumptions, but

$$c = \frac{(21/20)(9/10)}{11/10} = \frac{189}{220} > \frac{1}{10} = \alpha.$$

Thus the algebraic defect is confirmed, rather than merely an audit candidate.

Fix. Replace the comparison $c < \alpha$ by the valid bound

$$0 < c < 1 - \alpha < 1,$$

which follows from $\xi < \alpha$. The separate formal parameter derivation, `Tri.relaxedPaper_buffered_corner_real` and `Tri.relaxedPaper_room_and_corner` in `Tri/RelaxedTheorem3Paper.lean`, proves its buffered-corner inequalities without assuming $c < \alpha$. This confirms an error in an intermediate proof line, not a counterexample to Theorem 3; the numerical check alone does not verify every remaining step of the printed proof.

Erratum 12 (Lemma 10 — `finite size`). Location. *Natural Computing* 19 (2020), p. 262 (Springer PDF p. 14), Lemma 10 proof.

As printed. From $k \geq 4$ and $x < n - 2z$, the proof derives the displayed effective-rate inequality under the paper's informal "for n sufficiently large" convention.

Issue. Classification: (P) side condition. The finite statement is only asymptotically valid. A uniform sufficient condition is

$$\gamma \lg n \leq \frac{3n}{68}.$$

The common standing assumption $6\gamma \lg n \leq n$, equivalently $\gamma \lg n \leq 11.33n/68$, is weaker and therefore does not imply this Lemma 10 instance.

Fix. Carry the strict integer form $68\gamma \lg n + 1 \leq 3n$, or an equivalent explicit hypothesis, in every formal use of Lemma 10. The corrected Theorem 4 entry declaration `Tri.Byzantine.theorem4_entry` carries exactly this side condition.

Erratum 13 (Theorem 4 — `strictness`). Location. *Natural Computing* 19 (2020), p. 262 (Springer PDF p. 14), Theorem 4 proof immediately downstream of Lemma 10.

As printed. The proof treats $x \geq n - 2z$ as implying $y < z$.

Issue. Classification: (T). Since $x + y + z = n$, the inequality $x \geq n - 2z$ is equivalent to $y \leq z$, not to $y < z$.

Fix. Replace the strict inequality by $y \leq z$ and adjust the stopping predicate accordingly.

Erratum 14 (Lemmas 17–19 — `probability qualifiers`). Location. *Natural Computing* 19 (2020), pp. 266–267 (Springer PDF pp. 18–19): the displayed statements of Lemmas 17 and 18 on p. 266, and Lemma 19 on p. 267.

As printed. The displayed lemmas have no probability qualifier.

Issue. Classification: (T), but formalization-critical. Read literally, the statements are deterministic, which is false: adverse finite reaction or activation sequences have positive probability. The proofs and downstream uses are high-probability statements.

Fix. Attach the intended high-probability qualifier to each lemma, conditional on its stated start-of-stage hypotheses and inherited epidemic events:

- Lemma 17: from the stated range $n/(\gamma n \lg n)^{1/5} \le a < n/d^*$ and $E_Y(a) \le 14c^*\sqrt{\gamma a \lg n}$, the conclusion $E_Y(2a) \le 14c^*\sqrt{\gamma(2a)\lg n}$ holds with probability $1 - \exp(-\Omega(\gamma \lg n))$.
- Lemma 18: under the preceding Lemma 17 start-of-stage bound, the stage $a = n/d^* \to 2n/d^*$ creates active X-excess at least $2Q$, where $Q = \sqrt{\gamma n \lg n}$, with probability $1 - \exp(-\Omega(\gamma \lg n))$.
- Lemma 19: if the active X-excess is at least $2Q$ at $2n/d^*$ activations, then at full activation the active X-excess is at least Q with probability $1 - \exp(-\Omega(\gamma \lg n))$.

Erratum 15 (Lemma 18 — `stopping`). Location. *Natural Computing* 19 (2020), p. 266 (Springer PDF p. 18), Lemma 18 proof, in the sentence preceding estimates (i)–(iii) and in the subsequent stopped rate estimate.

As printed. With $Q = \sqrt{\gamma n \lg n}$, the proof asserts that if the excess of active reactions of type (2) over type (1) is at most $16Q$, then the stage completes without the active Y-excess ever exceeding $30Q$.

Issue. Classification: (P), correcting our own prior classification of this point as a non-erratum. Let E_t be active Y-excess and R_t be the running reaction excess. On the preceding high-probability events, $E_0 \le 14Q$ and the activation-prefix contribution is at most Q, so

$$E_t \le 15Q + 2R_t.$$

Crossing the $30Q$ barrier therefore forces only $R_t > 15Q/2 = 7.5Q$, not $R_t > 16Q$. For example, $R_t = 10Q \le 16Q$ permits $E_t = 35Q > 30Q$. The stopping-time phrasing in step (ii) correctly makes the local rate estimate non-circular, but it does not rule out exiting through the $30Q$ barrier before the claimed endpoint reaction-excess event.

Fix. Add a maximal stopped-walk bound

$$\max_{t \le \tau} R_t \le \frac{15Q}{2}$$

with failure $\exp(-\Omega(\gamma \lg n))$, where τ is the minimum of the stage end and the first $30Q$ crossing. The paper's own rate estimate and active-reaction-count bound supply this estimate. It is stronger than the endpoint bound needed later, since $15Q/2 < 16Q$.

Erratum 16 (Heavy-B — `exact emulation claims (P)`). Location. *Natural Computing* 19 (2020), p. 258 (Springer PDF p. 10), Sect. 4.1, "Correctness of the Heavy-B emulation," last paragraph.

As printed. With $\hat{x} = x + b$ and $\hat{y} = y + b$, the paper says that "the relative probability of reaction (1') to that of (2'), equals the relative probability of reaction (1) to that of (2) in the tri-molecular CRN," and concludes that the evolution of $\hat{x}, \hat{y}$ "exactly matches" that of x, y in Tri.

Issue. The informal verb "emulate" is not itself the error. The problem is the literal pair of claims used in the Section 4.1 correctness argument: the reaction odds are asserted to be equal, and the virtual process is asserted to match Tri exactly. Neither is true in general. This is a defect in the printed proof, not a refutation of the Heavy-B clause of Theorem 2.

Why. In a Heavy-B state (x, y, b), conditioned on a resolution reaction, the favorable and adverse propensities have ratio

$$xb : yb = x : y.$$

At the virtual Tri state $(\hat{x}, \hat{y}) = (x + b, y + b)$, the two productive tri-molecular propensities have ratio

$$\binom{\hat{x}}{2}\hat{y} : \hat{x}\binom{\hat{y}}{2} = (\hat{x} - 1) : (\hat{y} - 1) = (x + b - 1) : (y + b - 1).$$

Equality would require

$$x(y + b - 1) = y(x + b - 1),$$

or equivalently

$$(b - 1)(x - y) = 0.$$

Thus the odds agree only when $b = 1$ or $x = y$. For example, $(x, y, b) = (3, 1, 3)$ gives Heavy-B odds $3 : 1$, whereas the projected Tri odds are $5 : 3$. The raw interaction laws also use different entity populations: Heavy-B samples from $x + y + b$ entities, while the virtual population has $x + y + 2b$ entities. These two obstructions are machine-checked in `Tri/HeavyBEmulation.lean`: `Tri.heavy_tri_odds_agree_iff` proves the exact equality criterion, and `Tri.heavyToDouble_entity_count` proves the entity-count mismatch whenever $b > 0$.

Fix. Replace the exact-match argument by the one-sided comparison actually needed in the majority region: if $x \geq y$ and $b \geq 1$, the Heavy-B resolution skeleton is at least as biased toward X as productive Tri, with equality only in the two degenerate cases above. The proof must then combine this stochastic bias comparison with a Heavy-B-specific resolution clock; it cannot be obtained by black-box transport along an exact Markov lumping. The Heavy-B theorem itself is proved directly as `Tri.theorem2_heavyB`.

Erratum 17 (Single-B — `creation-imbalance concentration (P)`). Location. *Natural Computing* 19 (2020), p. 259 (Springer PDF p. 11), Sect. 4.4, Single-B phase-1 analysis, paragraph beginning "The probability that reaction $(0'_x)$ occurs is the same as the probability that reaction $(0'_y)$ occurs."

As printed. The paper observes that the two creation reactions $(0'_x)$ and $(0'_y)$ are equally likely and applies Lemma 2 to the number of $(0'_x)$ reactions among kn creation reactions. It then uses the resulting endpoint count to conclude that the virtual gap $\hat{x} - \hat{y}$ does not *become* smaller than the stated lower barrier during the stage.

Issue. Lemma 2 supplies concentration at a fixed endpoint, whereas the needed assertion is maximal: the creation-orientation imbalance must remain inside a barrier at every prefix before the stage terminates. An endpoint bound does not exclude a trajectory that crosses the adverse barrier and later returns. The printed argument also does not justify treating the history-dependent sequence of creation times as an independent sequence of trials.

Why. Let C_X and C_Y count the two creation orientations. Their masses are equal at every physical state, so $C_X - C_Y$ is an adapted fair walk when indexed by creation events. Fairness alone does not convert a terminal estimate into an ever-crossing estimate. The appropriate exponential process on the full seven-event chain is

$$G_t = \exp\left(\lambda(C_X - C_Y) - \frac{\lambda^2}{2}(C_X + C_Y)\right).$$

Inert and resolution reactions leave G_t unchanged. The two creation reactions multiply it by $\exp(\lambda - \lambda^2/2)$ and $\exp(-\lambda - \lambda^2/2)$ with equal mass, and

$$\frac{e^{\lambda} + e^{-\lambda}}{2} \le e^{\lambda^2/2}.$$

Hence G_t is a supermartingale without an independence assumption. If the number of creations is at most H, Ville's inequality gives

$$\Pr[\exists t:\ C_X(t) - C_Y(t) \ge D,\ C_X(t) + C_Y(t) \le H] \le \exp\left(-\frac{D^2}{2H}\right),$$

and the sign-reversed potential gives the opposite tail.

Fix. Replace the endpoint invocation of Lemma 2 by the stopped maximal estimate above, retain an explicit creation-count budget, and combine both creation tails with the corrected resolution-gap process before recovering the physical gap. The formalization proves the equal-mass input as `Tri.singleCompPMF_creation_masses_eq`, the full-kernel one-step supermartingale as `Tri.creationG_step`, the two maximal bounds as `Tri.creation_tail_maximal` and `Tri.creationY_tail_maximal`, and their optimized forms as `Tri.creation_tail` and `Tri.singleCreationY_tail`. The complete Single-B clause is `Tri.theorem2_singleB`.

Erratum 18 (Lemma 8 — `adaptive-policy comparison (P)`). Location. *Natural Computing* 19 (2020), p. 261 (Springer PDF p. 13), Lemma 8 and its proof in the Byzantine section.

As printed. For $0 < \ell \le n - x - z$, Lemma 8 says that

$$\pi^*((x, y, z), (x + \ell, y - \ell, z))$$

is minimized by the paper's fixed Byzantine response. The proof says it suffices to observe

$$\pi^*((x, y, z), q) \le \pi^*((x + 1, y - 1, z), q), \qquad q = (x + \ell, y - \ell, z),$$

because every path from the first state to q passes through the second.

Issue. The displayed inequality compares two starting states under one fixed transition law. The lemma's conclusion compares different Byzantine response policies, including responses that may depend on the complete past. Starting one step closer to a target under a fixed policy does not establish that one policy minimizes the target-hitting probability.

Why. Skip-freeness proves only the stated spatial monotonicity: for a fixed policy, every successful path from x to $x+\ell$ first visits $x+1$. A policy comparison needs a Bellman argument. For each finite horizon, the continuation value—the probability of reaching the upper target in the remaining steps—must first be proved monotone in the current X count. The pointwise worst-response inequality can then be applied to this monotone continuation value, followed by backward induction over the horizon and over the complete history. The eventual hitting comparison is obtained by taking the increasing supremum over finite horizons. Those policy-comparison steps are absent from the printed proof.

Fix. Replace the proof by finite-horizon dynamic programming and then pass to the unbounded hitting event. This repair is machine-checked as `Tri.Byzantine.lemma8`. Its finite-horizon core `lemma8_finite_horizon` proves Bellman domination against every history-dependent strategy; nearest-neighbour support identifies the upper Bellman target with the exact configuration, and a supremum over all finite horizons gives the paper's unbounded π^* comparison. The downstream potential form remains available as `controlledLaw_antitone_expect_le_pow` and `controlledLaw_antitone_super`.

Erratum 19 (Lemma 11 — `ratio inequalities (P)`). Location. *Natural Computing* 19 (2020), p. 263 (Springer PDF p. 15), Lemma 11 proof, the two-party and third-party backsliding calculations.

As printed. In the two-party calculation the paper sets $p = 1/2 + \Delta/(4n)$ and prints

$$\left(\frac{1-p}{p}\right)^{\Delta/8} < \left(1 - \frac{\Delta}{n}\right)^{\Delta/8}.$$

In the third-party calculation it sets $p = 1/2 + \Delta/(8n)$ and prints

$$\left(\frac{1-p}{p}\right)^{\Delta/4} < \left(1 - \frac{\Delta}{2n}\right)^{\Delta/4}.$$

Issue. Both displayed comparisons are false: before taking powers, the left-hand ratio is strictly *larger* than the printed linear factor. These two displayed comparisons are the entirety of the "false ratio inequalities" defect. No stronger defect in Lemma 11 is asserted: the paper's two-mode decomposition is legitimate, and its stated $m\exp(-\Omega(\Delta^2/n))$ conclusion survives after repairing these two estimates.

Why. In the two-party case,

$$\frac{1-p}{p} = \frac{2n-\Delta}{2n+\Delta},$$

and

$$\frac{2n-\Delta}{2n+\Delta} - \left(1 - \frac{\Delta}{n}\right) = \frac{\Delta^2}{n(2n+\Delta)} > 0.$$

For example, $n = 100$, $\Delta = 10$ gives $190/210 \approx 0.90476 > 0.9$. In the third-party case,

$$\frac{1-p}{p} = \frac{4n-\Delta}{4n+\Delta},$$

and

$$\frac{4n-\Delta}{4n+\Delta} - \left(1 - \frac{\Delta}{2n}\right) = \frac{\Delta^2}{2n(4n+\Delta)} > 0.$$

For $n = 100$, $\Delta = 10$ this is $390/410 \approx 0.95122 > 0.95$.

Fix. Use the valid inequality

$$\frac{1-t}{1+t} \le e^{-2t} \qquad (0 \le t < 1).$$

With $t = \Delta/(2n)$ in the two-party case,

$$\left(\frac{1-p}{p}\right)^{\Delta/8} \le \exp\left(-\frac{\Delta^2}{8n}\right).$$

With $t = \Delta/(4n)$ in the third-party case,

$$\left(\frac{1-p}{p}\right)^{\Delta/4} \le \exp\left(-\frac{\Delta^2}{8n}\right).$$

Union over the two modes and the $m-1$ competitors preserves the paper's asymptotic form $m\exp(-\Omega(\Delta^2/n))$. The paper's mode bounds are now machine-checked for the cumulative mode counters. To identify the certified events, write D for the formalization's integer initial gap lower bound, corresponding to Δ, and assume $n \ge 3$, $4 \le D \le n$, and initial gap at least D against every competitor.

Let $A_Y(t)$ and $H_Y(t)$ be the cumulative direct and third-party contributions to the X–Y gap, respectively, both initially zero. A productive reaction between X and Y contributes ± 2 to A_Y; a productive reaction between one of X, Y and a third species contributes ± 1 to H_Y. Other reactions contribute zero to both counters. The third-party counter is aggregated over all third species. Their sum is the change in the actual X–Y gap. The two bad events on the unrestricted productive chain are

$$B_2 = \{\exists\, Y \ne X,\ t:\ A_Y(t) \le -D/4\}, \qquad B_3 = \{\exists\, Y \ne X,\ t:\ H_Y(t) \le -D/4\}.$$

Stopping the proof potentials at the first crossing of either counter for any competitor gives the sharp odds bound

$$\Pr(B_2 \cup B_3) \le (m-1)\left[\left(\frac{2n-D}{2n+D}\right)^{\lceil D/8\rceil} + \left(\frac{4n-D}{4n+D}\right)^{\lceil D/4\rceil}\right] \le 2(m-1)\exp\left(-\frac{D^2}{8n}\right).$$

Combining this with the probability ceiling 1 yields

$$\Pr(B_i) \le \Pr(B_2 \cup B_3) \le m\exp\left(-\frac{D^2}{18n}\right), \qquad i = 2, 3.$$

The sharp union estimate is `Tri.Multi.productivePairModeEverMass_le_sharpOdds`; the two paper-facing wrappers are `Tri.Multi.lemma11_twoPartyBacksliding_modes` and `Tri.Multi.lemma11_thirdPartyBacksliding_modes` in `Tri/PaperLemma11ModeBounds.lean`. The same file proves the corresponding finite-horizon bounds, without a horizon factor.

The older declarations `Tri.Multi.lemma11_twoPartyBacksliding` and `Tri.Multi.lemma11_thirdPartyBacksliding` in `Tri/PaperLemma11.lean` both bound the global actual half-gap event `productivePairGapEverFailureh3X(D/2)c0`, via `Tri.Multi.productivePairGapEver_half_failure_exp`. They do not themselves certify the separate cumulative mode events.

Non-errata: differences that are ours, not the paper's

For the record, the following divergences between the Lean development and the paper are *our* choices or *our* earlier mistakes, and are deliberately not listed above.

- Phase 0 of Theorem 1(a) is proved with a concave symmetric Lyapunov supermartingale rather than the paper's foray/checkpoint combinatorics. Both are correct; ours was easier to formalize.
- Corollary 3's proof omits a monotonicity step: the productive-clock estimate may be proved at the boundary minority size $y = \gamma \lg n$ and then pushed to starts with smaller minority. This is a true omitted step, not an erratum, because the statement remains correct. The formalization records the missing comparison as `Tri.corollary3_boundary_success_le_entry_success`, built from the reusable Bellman comparison `Tri.productive_hitProb_mono_start_of_upper`.
- An early version of our Corollary 3 wrapper stated a raw-interaction result under the productive-clock name. That was our labelling error, now corrected, and is not a defect of the paper.
- Lemma 18's deterministic endpoint ledger closes with *zero* slack. The paper's own arithmetic is $48\sqrt{\gamma n \lg n} - 14\sqrt{\gamma n \lg n} - 32\sqrt{\gamma n \lg n} = 2\sqrt{\gamma n \lg n}$, exactly the required gap. A formalization has no room to lose a constant anywhere in this lemma.